

Halfway Escape Optimization: A Quantum-Inspired Solution for General Optimization Problems

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Abstract

This paper first proposes the Halfway Escape Optimization (HEO) algorithm, a quantum-inspired metaheuristic designed to address general optimization problems. The HEO mimics the effects between quantum such as tunneling, entanglement. After the introduction to the HEO mechanisms, the study presents a comprehensive evaluation of HEO's performance against extensively-used optimization algorithms, including Particle Swarm Optimization (PSO), Genetic Algorithm (GA), Artificial Fish Swarm Algorithm (AFSA), Grey Wolf Optimizer (GWO), and Quantum behaved Particle Swarm Optimization (QPSO). The primary analysis encompasses 14 benchmark functions with dimension 30, demonstrating HEO's effectiveness and adaptability in navigating general optimization problems. The test of HEO in Pressure Vessel Design and Tubular Column Design also infers its feasibility and potential in real-time applications. Further validation of HEO in Osmancik-97 and Cammeo Rice Classification achieves a higher accuracy record.

Keywords: Swarm intelligence, Optimization, Metaheuristics, Halfway Escape Optimization(HEO)

Highlights

- we propose a new algorithm called HEO
- we provide a comprehensive benchmark of 14 Objective Functions for our HEO algorithm
- we further evaluate the effectiveness of HEO in 3 classical engineering problems
- we optimize Logistic Regression model with HEO, achieving a higher record in Osmancik-97 and Cammeo Rice Classification

1 Introduction

The main motivation behind the development of the Halfway Escape Optimization (HEO) algorithm stems from the limitations of existing optimization methods in terms of efficiency and adaptability to various industrial single-objective optimization problems.

HEO is a quantum-inspired metaheuristic designed to tackle the distinction between different search spaces in industrial scenarios with a focus on achieving fast convergence rates and reducing the time cost in searching. Inspired by the behavior of quantum particles and the concept of halfway escape, the algorithm offers a versatile and adaptive approach to exploration and exploitation in challenging optimization domains.

This paper provides a comprehensive analysis to the HEO algorithm, focusing on its core principles, adaptability, and robustness in navigating diverse optimization landscapes. The unique energy-driven behavior, vibration strategies, and exploratory mechanisms of HEO are examined in the context of solving a range of simple as well as complex benchmark functions. The evaluations demonstrate HEO's effectiveness in achieving convergence, balancing exploration and exploitation, and discovering high-quality solutions across a variety of challenging optimization problems.

Comparative studies with other swarm optimization algorithms highlight HEO's adaptability and robustness in addressing the complexities of multimodal and high-dimensional search spaces, showcasing its fast convergence speed. The results of this study underscore the importance of the HEO algorithm as a promising approach to address various optimization challenges, with implications for a wide range of practical applications, including structural design [Elegbede \(2005\)](#) [ELGebaly \(2019\)](#) and model optimization [Wang et al \(2019\)](#).

The detailed analysis presented demonstrates the potential of the HEO algorithm as a powerful and adaptive optimization method. Overall, the paper first proposes HEO method and primary testing its effectiveness in 14 single objective functions as well as 3 engineering problems including Pressure Vessel Design, Tubular Column Design, Osmancik-97 and Cammeo Rice Classification.

2 Related Work

In the realm of optimization, metaheuristic algorithms play a key role in the addressing of challenging optimization problems. These algorithms draw inspiration from natural phenomena and collective behaviors to navigate high-dimensional and multimodal search spaces, offering adaptive and robust solutions and widely using in real-world optimization problems such as scheduling or logistics problems [Obagbuwa \(2018\)](#).

In 1975, [Holland](#) created a new metaheuristic algorithm called Genetic Algorithm (GA), a prominent evolutionary algorithm, that simulates natural selection and genetic recombination to evolve a population of candidate solutions. GA's adaptability has made it a go-to choice for a wide range of optimization problems. However, its exploration of high-dimensional and multimodal search spaces may be limited, potentially leading to suboptimal solutions.

Particle Swarm Optimization (PSO) has gained widespread recognition as a bio-inspired metaheuristic algorithm rooted in the collective behavior of bird flocks. PSO was first proposed by [Eberhart and Kennedy](#) in 1995. PSO iteratively updates the positions and velocities of the particles based on their best individual and global solutions, enabling effective exploration and exploitation of the search space([Eberhart and Kennedy, 1995](#)). While PSO has exhibited robustness in addressing various optimization problems, its potential for premature convergence and challenges in handling various multimodal landscapes have been noted([Vázquez et al, 2013](#)).

Ant Colony Optimization (ACO) is an algorithm inspired by the foraging behavior of ants. It was initially proposed by Marco Dorigo in his doctoral thesis in 1992 and has since gained significant attention in the field of optimization([Dorigo et al, 1996](#)). ACO simulates the behavior of ants in finding the shortest path between their nest and food source by depositing pheromone trails. These trails act as a form of communication, allowing ants to navigate and explore the search space effectively. ACO has been successfully applied to a wide range of optimization problems, including the well-known Traveling Salesman Problem (TSP)[Tanakitkorn et al \(2014\)](#) and Vehicle Routing Problem (VRP)([Dorigo et al, 1996](#)). Its ability to find near-optimal solutions and its adaptability to dynamic environments make ACO a valuable algorithm for solving complex optimization problems.

Differential Evolution (DE) is another population-based stochastic optimization algorithm that was introduced by [Storn and Price \(1997\)](#). DE operates by iteratively evolving a population of candidate solutions through a combination of mutation, crossover, and selection operations. It is inspired by the process of natural evolution and survival of the fittest. DE utilizes the differential operators to explore and exploit the search space efficiently. It has been widely applied to various optimization problems and has shown promising performance. The simplicity, effectiveness, and robustness of DE make it a popular choice for solving real-world optimization problems.

Artificial Fish Swarm Algorithm (AFSA) emulates the foraging behavior of fish in a swarm to explore the search space and update candidate solutions([Li et al, 2002](#)). AFSA has demonstrated effectiveness in continuous and combinatorial optimization tasks by adopting swarm and follow behaviors, yet its ability to search may present challenges due to its reliance on fish movement and interaction dynamics.

The Firefly Algorithm (FA) is a nature-inspired optimization algorithm proposed by [Yang](#) in 2009. Drawing inspiration from the flashing patterns of fireflies, FA simulates the social behavior of fireflies to solve optimization problems. Each firefly's light emission represents a potential solution, and the attractiveness between fireflies is determined by the brightness of their light and their distance from each other. Fireflies seek to improve their positions by moving towards brighter fireflies in the search space. FA has demonstrated effectiveness in solving a variety of optimization problems, including continuous, discrete, and multi-modal functions.

In 2009, Quantum behaved Particle Swarm Optimization (QPSO), or Q-PSO, was proposed to solving multi-objective problem [Omkar et al \(2009\)](#). QPSO is a variant of Particle Swarm Optimization (PSO) that incorporates concepts from quantum mechanics, such as quantum probabilities and superposition, to enhance its search capabilities. It aims to find optimal solutions for multi-objective problems by leveraging the principles of quantum mechanics and the collaborative behavior of a swarm of particles.

The Grey Wolf Optimizer (GWO) draws inspiration from the hierarchical structure and hunting behavior of grey wolves in nature, which suggested by [Mirjalili et al](#) in 2014. Through the alpha, beta, and delta wolf concept, GWO dynamically updates solution positions, showcasing promise in addressing complex optimization problems. Nonetheless, the algorithm's sensitivity to parameter settings and the need for domain-specific fine-tuning have been identified as areas for consideration [Mirjalili et al](#).

The Salp Swarm Algorithm (SSA) draws inspiration from the collective behavior of salps in the ocean ([Mirjalili et al, 2017](#)). This bio-inspired optimization technique emulates the swarming and foraging dynamics of salps to navigate complex search spaces and seek optimal solutions. By simulating the intricate swimming and feeding patterns of salps, SSA exhibits a remarkable capacity for exploration and exploitation, making it proficient in addressing a wide range of optimization problems, including both unconstrained and constrained scenarios.

There still have other meta-heuristic algorithms, those algorithms could be classified as two categories: bio-inspired and the nature-inspired, nature-inspired means the optimization algorithms are inspired from physical or chemical laws in nature. Bio-inspired optimization algorithms including Whale Optimization Algorithm (WOA) [Mirjalili and Lewis \(2016\)](#), Artificial Bee Colony (ABC) [Kang et al \(2011\)](#), Bat Algorithm (BA) [Yang \(2010\)](#), Cuckoo Search (CS) [Yang and Deb \(2009\)](#), Dragonfly Algorithm (DA) [Mirjalili \(2015\)](#), Flower Pollination Algorithm (FPA) [Yang \(2012\)](#), Cheetah Optimizer (CO) [Akbari et al \(2022\)](#), Genetic Programming (GP) [Willis \(1997\)](#), Fox Optimizer (FOX) [Mohammed and Rashid \(2022\)](#), GOOSE algorithm [Hamad and Rashid \(2024\)](#).

Nature-inspired optimization algorithms contain Snow Ablation Optimizer (SAO) [Deng and Liu \(2023\)](#), Light Spectrum Optimization (LSO) [Abdel-Basset et al \(2022\)](#), Nuclear Reaction Optimization (NRO) [Wei et al \(2019\)](#), Water Cycle Algorithm (WCA) [Eskandar et al \(2012\)](#), Chernobyl Disaster Optimizer (CDO) [Shehadeh \(2023\)](#). Those bio-inspired as well as nature-inspired algorithms both have contributed significantly to the field of optimization, each with its unique strengths.

3 Proposed Methodology

In recent years, swarm optimization algorithms have been widely applied in various fields, such as machine learning, engineering design. These algorithms have achieved remarkable success in solving optimization problems and have shown great potential in handling complex tasks.

However, these algorithms often face difficulties when faced with complex landscapes. One example of a group optimization algorithm facing a rugged landscape is the Particle Swarm Optimization (PSO) algorithm. When facing rugged landscapes with many local optima, PSO often gets trapped in these local optima and fails to find the global optimum [Vázquez et al \(2013\)](#).

The difficulty in finding global optima in rugged landscapes can be attributed to several reasons. First, the rugged landscape creates many false peaks, making it challenging for the optimization algorithm to distinguish between the local optima and the global optimum. Second, the steep gradients in rugged landscapes lead to rapid convergence to local optima, which hinders the exploration of the search space [Han et al \(2018\)](#). Third, the presence of many local optima increases the probability of particles getting trapped in these optima. Especially in high-dimensional situations, those hollow spaces would further reduce the chances of finding the global optimum [Cai et al \(2020\)](#).

To solve those issues, the paper designed HEO (Halfway Escape Optimization) based on a few behaviors from quantum. HEO algorithm includes four different mechanisms for exploitation ([3.4 Center Clipping](#)) as well as exploration ([3.2 Position Update](#), [3.3 Vibration](#), [3.5 Random Skip](#)).

Yet QPSO and HEO are both quantum-inspired algorithms with the same mechanism in computing global weight for updating position (seen in equation (6)), they still have significant differences. In contrast to the QPSO, QPSO only allows for a single state change, HEO is more inclined to escape when convergence is achieved, resulting in the inclusion of velocity during the escape phase. Moreover, HEO incorporates random searches not only within the range of updates but also for the entire swarm. Additionally, utilizing a center clipping strategy in HEO contributes to its superior convergence rate.

The Position Update mechanism, which forms the basis of all swarm optimization algorithms, which are the moving way of each individual entity in the swarm, is designed differently in HEO. Instead of focusing on a single peak, the quantum in HEO has two states that determine whether it should approximate the optimum. This unique way of updating positions allows the swarm in HEO to escape from false peaks, thus addressing the first difficulty mentioned earlier. Like the quantum could shuttle through the potential energy barrier [Razavy \(2003\)](#), the movement of the quantum in HEO is not influenced by gradients like in the PSO algorithm, making it more stable and less likely to be hindered by complex landscapes. Furthermore, in order to navigate complex hollow landscapes, HEO incorporates mechanisms such as increasing the escape factor, Vibration, and Random Skip to accelerate the quantum's ability to overcome obstacles. Lastly, for exploring complex situations, the Center Clipping mechanism in HEO ensures that the swarm is relatively concentrated around

the already discovered optima, thereby efficiently searching for solutions in a convex environment.

3.1 HEO Overview

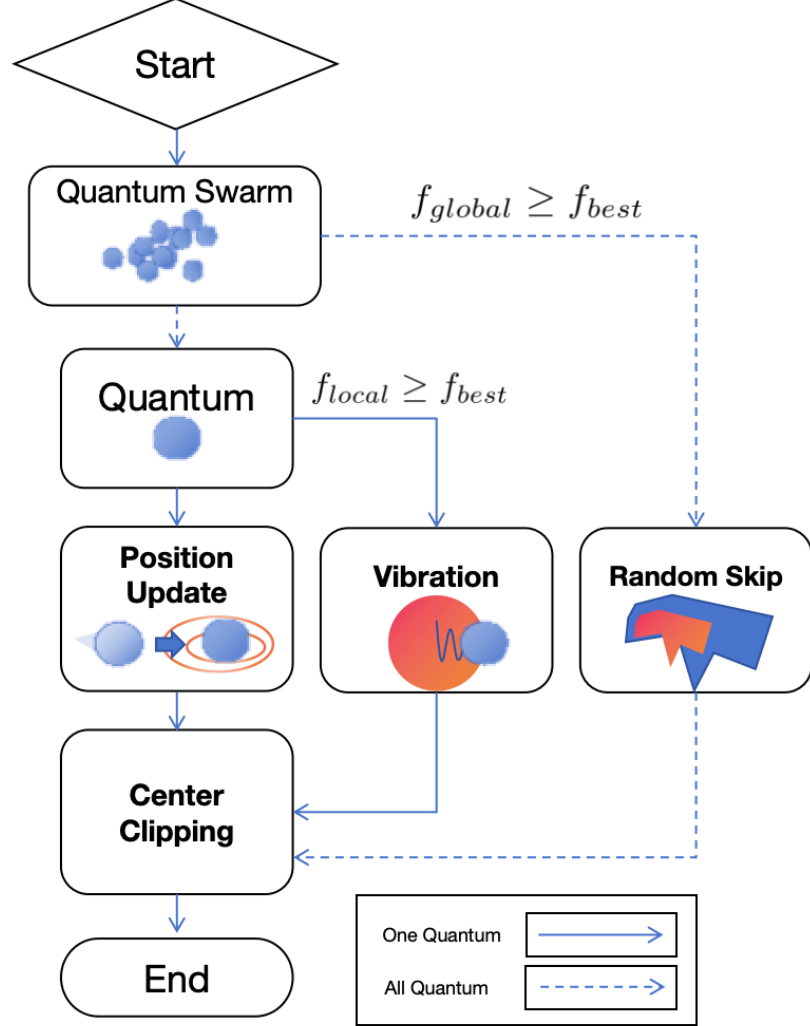


Fig. 1 Simplified Flowchart of One Iteration in HEO

As Fig.1 shown, in the Halfway Escape Optimization (HEO) algorithm, the swarm is structured as a collection of quantum entities. Each quantum entity iteratively adjusts its position and employs the Center Clipping strategy to satisfy the defined constraints while enhancing the efficiency of the search process. Should the swarm fail to identify

a superior global optimum, a random skip is initiated, allowing for a departure from the current trajectory. Concurrently, individual quantum entities engage in a vibration mechanism when local optima are not improved. This action aims to explore beyond the plateaus or local convex hulls, facilitating a more comprehensive exploration of the search space. Details of HEO can be seen in **Algorithm 1**.

Algorithm 1 Halfway Escape Optimization (HEO)

```

procedure HEO( $k, i_{max}, a_{max}, c_{max}, f, p, bound, R$ )
   $Q \leftarrow$  list of  $k$  quantums with positions, levels and local optimas
   $f_{best} \leftarrow \min(\{f(q.x) \mid q \in Q\})$ ,  $c_i \leftarrow 0$ 
   $q_{best} \leftarrow$  quantum in  $Q$  with fitness  $f_{best}$ 
  for  $i \leftarrow 1$  to  $i_{max}$  do
    for all  $q$  in  $Q$  do
      update position of  $q$ 
       $f_q \leftarrow f(q.x)$ 
      if  $f_q < f_{best}$  then
        Update  $f_{best}, q_{best}$ 
         $c_i \leftarrow \text{int}(c_i/2)$ 
      else if  $f_q < f(q.local\_best.x)$  then
        Update  $q.local\_best$ 
         $q.a_i \leftarrow \text{int}(q.a_i/2)$ 
      else
         $n \leftarrow N(0, \sigma_{q.x}, p)$ 
         $q.x \leftarrow q.x + \frac{n}{1+e^{a_i}}$ 
      end if
      center clipping  $q.x$  in  $S_{result}$ 
    end for
    if  $q.a_k * r_4 < \frac{a_{max}-1}{2}$  then
       $q.a_k \leftarrow q.a_k + 1$ 
    end if
    if  $c_i > c_{max}$  then
      every quantum  $q$  random skip
       $c_i \leftarrow 0$ 
    end if
     $c_i \leftarrow c_i + 1$ 
  end for
  return  $f_{best}, q_{best}$ 
end procedure

```

3.2 Position Update

The mechanism Position Update of HEO, as it is named, is the special way that quantum updates its position for search, which is the fundamentals of all optimization algorithms. Similar to PSO, in HEO, the updates of quantum's position consider both

the local optimum and global optimum. The core concept of the HEO is "escape", the quantum in HEO would move towards the opposite direction once the group gets stuck in the local optima, this trait allows HEO to escape from it. First of all, as Fig.2 shows, the HEO is not fully gradient dependent like PSO, HEO only uses fitness or cost for updating the velocity of escape as well as updating the bound of the quantum, which makes HEO find relatively good solutions on the sheer and other narrow landscapes.

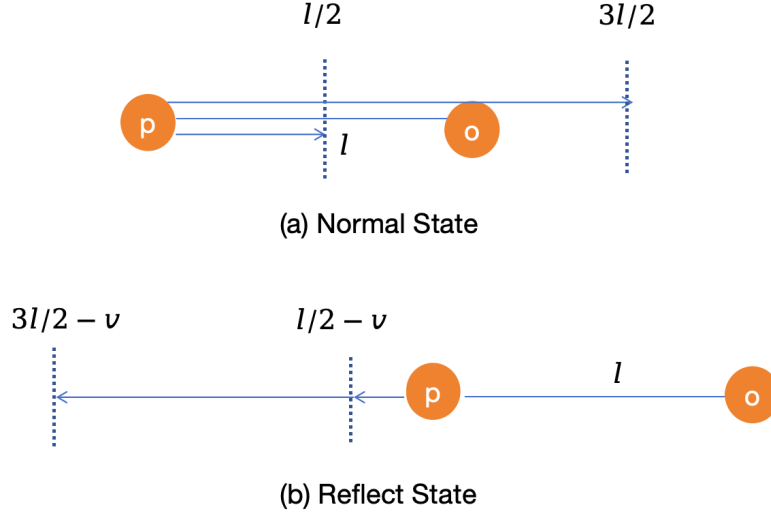


Fig. 2 Position Update in HEO

$$\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{v}_g + \mathbf{v}_1, \quad i = 1, 2, 3 \dots N \quad (1)$$

$$\mathbf{v}_g = (\mathbf{x}_g - \mathbf{x}_i(c_i + 1) \times r_1) \times r_2 \times r_3 \quad (2)$$

$$\mathbf{v}_1 = (\mathbf{x}_1 - \mathbf{x}_i(c_i + 1) \times r_1) \times r_2(1 - r_3) \quad (3)$$

In the normal state(a), quantum does not have any velocity. When the escape factor c_i is equal to zero the behavior is equivalent to finding the solution surrounding the halfway of optima in equation (1). To be precise, the HEO algorithm would get into reflect state(b) as Fig.2 shows, to find other possible solution spaces. Unlike directly searching randomly, the reflecting state of the HEO's quantum utilizes the information(which is c_i here) from local optima, allowing it to escape from it faster.

$$r_1 \sim U(1 - R, 1 + R) \quad (4)$$

$$r_2 \sim U(0.5, 1.5) \quad (5)$$

$$r_3 \sim U(0, 1) \quad (6)$$

$$c_{i+1} = \begin{cases} 0 & (\text{when } i = 0) \\ c_i + 1 & (\text{when } f_{global} > f_{best}) \\ c_i \mid 2 & \text{else} \end{cases} \quad (7)$$

As for the random factors r_1 and r_2 shown in equation (4) and equation (5), r_1 is used to control the magnitude of an escape step, making the HEO escape behaviors more efficient than the simple grid search. r_2 is added to further randomize the process. Both of those two random variables have exception 1 to ensure the feasibility. The random factor r_3 controls the global weight of this update process, creating a fan area to search for the solution.

3.3 Vibration

Another behavior of HEO is common in other algorithms, including AFSA, when the search is meaningless to make a random step for an individual search. Yet those algorithms adopt the strategy, they ignore the influence of size of steps, and those search approaches highly rely on the hyper-parameters tuning. HEO tries to use the standard deviation of positions to judge the magnitude of one random step, creating a relatively stable performance compared to the others, this random search is called Vibration (seen in equation (8)).

$$\mathbf{x}_{i+1} = \mathbf{x}_i + \frac{\mathbf{n}}{1 + e^{a_i}} \quad (8)$$

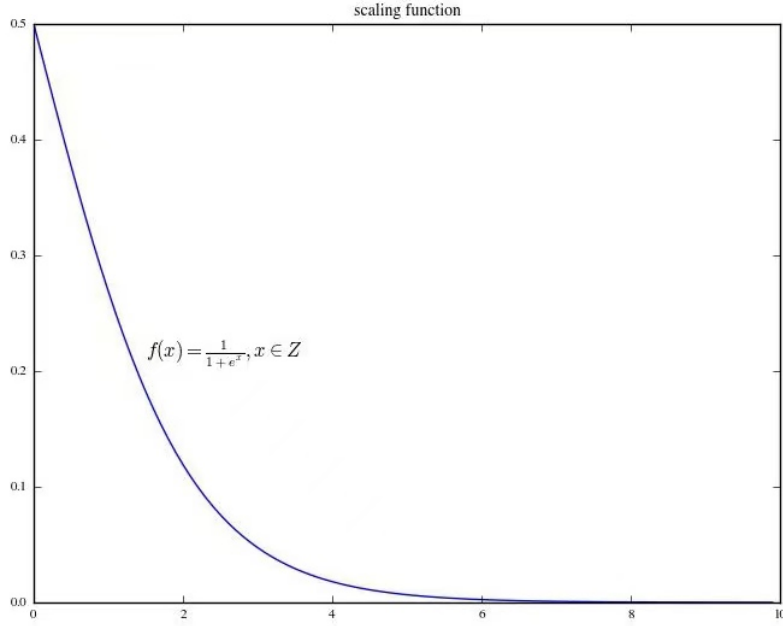


Fig. 3 Scaling Function

$$a_{i+1} = \begin{cases} a_i + 1 & (\text{when } a_i * r_4 < a_{max}) \\ a_i \mid 2 & (\text{when } f_{local} < f_{best}) \end{cases} \quad (9)$$

$$r_4 \sim U(0, 1) \quad (10)$$

$$\mathbf{n} \sim N(0, \sigma_{x_i}) \quad (11)$$

Just like the property of quantum in physics, with higher energy level a_i , the quantum would be harder get influenced by others(Sommerfeld, 1916), which means smaller oscillations(seen in equation (8) and Fig.3), maintain an extent of diversity in the population. To maintain the convergence of the whole group, the increment of energy level a_i only works in a few quantum. As for amplitude of vibration, other scaling approaches to adapting the space of high dimensions with gamma functions, are computed much slower than just simply calculating the standard deviation of the position vectors as shown in equation(11). The reason for using Normal distribution due its universal applicability. According to the central limit theorem, with the increasing of the dimension, the size of the vibration is more approximate to the size of the step in the real situation. Since the HEO does not include periodical functions in this behavior, it cannot be defined as oscillations, that's the reason for its name.

3.4 Center Clipping

Just like quantum entanglement causes quantum attraction, so the distance of quantum escape is finite, the quantum group in HEO would restrict the individuals, ensuring them in a certain area to avoid over-escape by the behavior.

$$S_{result} = S_{bound} \cap S_{group} \quad (12)$$

$$b_g = \|\mathbf{x}_i - \mathbf{x}_g\|_2 \times r_5 \quad (13)$$

$$r_5 \sim U(0, 2) \quad (14)$$

As Fig.4 illustrated, the quantum not only would be restricted by the search area or constraints S_{bound} (the area with (a)) with half of the side length b , but the group also would be restricted by S_{group} . S_{group} (the area of (b) in Fig.3)is the search space of the group that HEO allows to search with half of the side length b_g , the search space in HEO are all in the shape of a square, making it computing faster than calculating sphere. Moreover, as equation (13) shows, this bound of group S_{group} depends on the differences between the current quantum's position and the best one that HEO found. This center clipping mechanism uses a random factor r_5 , which makes this bound soft edge not converge too early and maintains the diversity of the group. The final result of the group gets into the range of S_{result} (area (c)), which is the overlap area of the S_{group} (area (b)) and S_{bound} (area (a)).

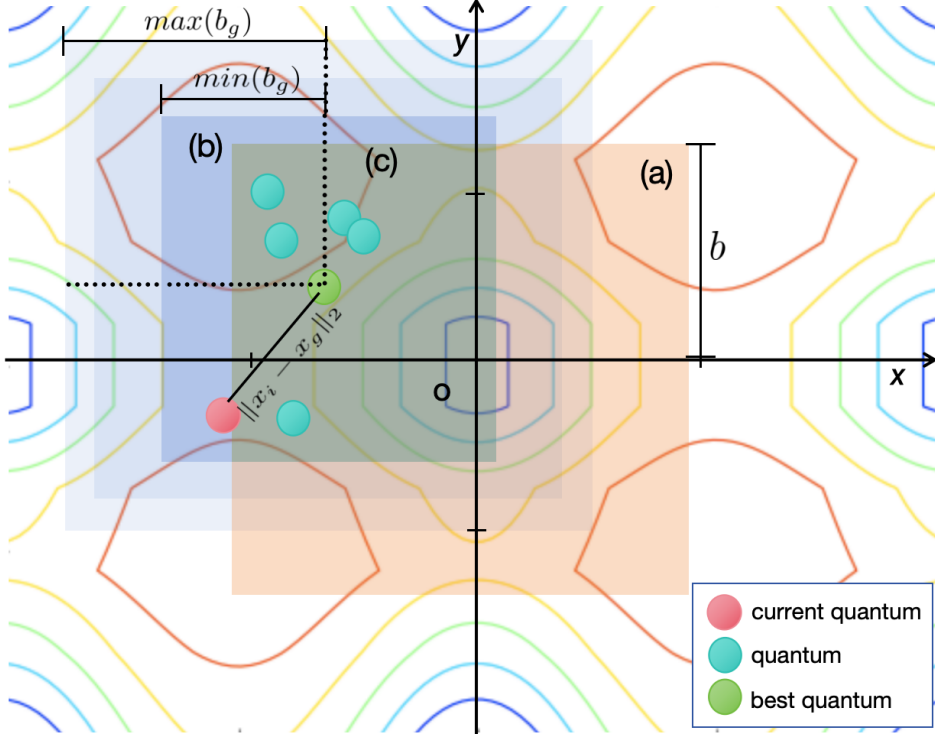


Fig. 4 Center Clipping in Ackley Function

3.5 Random Skip

The HEO is a swarm algorithm extremely focused on exploration. Except for the vibration behavior that each quantum has, the group itself would be randomly skipped to search the solution with the range of the bound. In equation (14), once the count c_i reached c_{max} and then manually defined, the whole group would make this random skip for spanning a totally new space. The reason to use this skipping mechanism is the higher randomness compared to using any periodical function to search in HEO.

$$\mathbf{x}_i = \frac{\mathbf{x}_i + \tilde{\mathbf{r}}}{2} \quad (15)$$

$$\tilde{\mathbf{r}}_j \in \tilde{\mathbf{r}} \quad (16)$$

$$\tilde{\mathbf{r}}_j \sim U(0, \mathbf{b}) \quad (17)$$

The random skip behavior in HEO allows the entire group of quantum to explore a completely new and random search space, the blue area represents the search space, and the orange part is for the constraint. This is achieved by randomly shifting the position of each quantum by a random vector $\tilde{\mathbf{r}}$ in Equation (15). The components of the random vector $\tilde{\mathbf{r}}_j$ are uniformly sampled from the interval $[0, b]$, as shown in equation (17). There are numerous b for $\tilde{\mathbf{r}}$, hence the shape of the search space also

changed with it, also the space restricts the constraints. This random skip mechanism enables HEO to explore different regions of the search space and potentially discover new and better solutions.

4 Proposed Extensions to HEO

This section describes practical extensions we implemented on top of the original HEO algorithm to improve numerical stability and convergence while preserving the escape-based search behavior. We refer to the baseline algorithm in ? as **Original HEO**.

4.1 v2: Stabilized Position Update (Removing the r_2 Term)

In the original HEO position update, the random factor r_2 acts as an additional stochastic amplification term. In our reproduction experiments, we observed that this amplification increases variance of the update magnitude and can destabilize the search. Therefore, in our **v2** variant we remove r_2 from all position-update equations while keeping the remaining stochastic factors and the escape counter mechanism unchanged.

Let x_i^t be the position of particle i at iteration t , x_g^t the global best position, and $x_{\ell,i}^t$ the personal best position. With an escape-dependent multiplier m_i^t (defined below), the v2 update is:

$$x_i^{t+1} = x_i^t + v_{g,i}^t + v_{\ell,i}^t, \quad (1)$$

$$v_{g,i}^t = (x_g^t - x_i^t \cdot m_i^t \cdot r_1) \cdot r_3, \quad (2)$$

$$v_{\ell,i}^t = (x_{\ell,i}^t - x_i^t \cdot m_i^t \cdot r_1) \cdot (1 - r_3), \quad (3)$$

where $r_1 \sim \mathcal{U}(1 - R, 1 + R)$ and $r_3 \sim \mathcal{U}(0, 1)$ are applied component-wise. To prevent uncontrolled growth when stagnation increases, we cap the escape multiplier using:

$$m_i^t = \min(1 + \log(1 + c_i^t), m_{\max}), \quad (4)$$

where c_i^t is the stagnation/escape counter for particle i .

4.2 v2: Convergent (Annealed) Vibration

The original vibration/noise mechanism can introduce persistent randomness that prevents late-stage contraction. We replace it with a bounded, annealed vibration term:

$$x_i^{t+1} \leftarrow x_i^{t+1} + \sigma(t) \epsilon, \quad \epsilon \sim \mathcal{N}(0, \Sigma), \quad (5)$$

where $\sigma(t)$ decays to zero as t approaches the maximum number of iterations T . In our implementation we use an exponential schedule

$$\sigma(t) = \sigma_0 \exp\left(-\frac{t}{T}\right), \quad (6)$$

so that vibration supports exploration early and allows convergence in later iterations.

4.3 v2: ES Local Search with the 1/5 Success Rule

To strengthen exploitation, we add a $(1+1)$ evolution-strategy (ES) local search step after the main v2 update. A mutated candidate is generated as:

$$x'_i = x_i + \sigma_i \eta, \quad \eta \sim \mathcal{N}(0, I). \quad (7)$$

If $f(x'_i) < f(x_i)$ the mutation is accepted. The step size σ_i is adapted using the 1/5 success rule over a sliding window of trials:

$$\sigma_i \leftarrow \begin{cases} \sigma_i / c, & \text{if success rate} > 0.2, \\ \sigma_i \cdot c, & \text{otherwise,} \end{cases} \quad (8)$$

where $c \in (0, 1)$ is a contraction factor.

4.4 CIBA: Affinity-Controlled Mutation

We further experimented with a mutation operator inspired by immune-system affinity mechanisms. The mutation magnitude is controlled by an affinity measure relative to the best fitness in the population:

$$\text{aff}(x_i^t) = f(x_i^t) - f(x_{best}^t), \quad (9)$$

and the update is:

$$x_i^{t+1} = x_i^t + r \cdot \beta \cdot \exp(-\|\text{aff}(x_i^t)\|), \quad (10)$$

where r is a random vector and β controls the mutation scale. This mechanism increases mutation for poor solutions and reduces it near high-quality solutions.

4.5 DirCIBA: Directional Affinity-Controlled Mutation

Basic CIBA controls mutation magnitude but does not encode a preferred direction, which can lead to random-walk behavior near the optimum. We therefore introduce **Directional CIBA (DirCIBA)**, where the mutation direction is a mixture of a global-best direction and a random direction:

$$\Delta x_i^t = \Delta_i^t \left(\alpha \cdot \frac{x_{best}^t - x_i^t}{\|x_{best}^t - x_i^t\| + \epsilon} + (1 - \alpha) \cdot \eta \right), \quad \eta \sim \mathcal{N}(0, I), \quad (11)$$

with mutation amplitude

$$\Delta_i^t = \beta \left(1 - \exp(-k \cdot \max(0, \text{aff}(x_i^t))) \right). \quad (12)$$

Here $\alpha \in [0, 1]$ controls exploitation, k controls sensitivity to affinity, and ϵ is a small constant for numerical stability.

5 Experimental Results

5.1 Experiment Design

The paper selects 14 benchmark functions with a relatively large bound within -100 to 100 (the reason to set a same bound is to test the adaptability of each algorithms) from famous objective functions that have extensional forms in high dimensions, including 7 unimodal functions in TABLE 1 and 7 multimodal functions in TABLE 2. The experiment would test the performance of classical metaheuristic algorithms including PSO, AFSA, GWO, and GA with the HEO algorithm (check the standard HEO in **Algorithm 1**) in dimension $p = 30$, and searching within 1000 iterations. Each swarm of the testing algorithms have 100 entities, which is common size of the swarm for testing as a rule of thumb. In addition, the experiment is validated 30 times to remove the potential impact caused by randomness.

Table 1 Test Unimodal Functions

Name	Function	range	f_{min}
Sphere	$F_1(x) = \sum_{i=1}^n x_i^2$	[-100,100]	0
Step	$F_2(x) = \sum_{i=1}^n (x_i + 0.5)^2$	[-100,100]	0
Schwefel 2.21	$F_3(x) = \max_i x_i $	[-100,100]	0
Schwefel 2.22	$F_4(x) = \sum_{i=1}^n x_i + \prod_{i=1}^n x_i $	[-100,100]	0
Rosenbrock	$F_5(x) = \sum_{i=1}^{n-1} [(x_{i+1} - x_i^2)^2 + (x_i - 1)^2]$	[-100,100]	0
BentCigar	$F_6(x) = x_1^2 + 10^6 \sum_{i=2}^n x_i^2$	[-100,100]	0
Sumsquares2	$F_7(x) = \sum_{i=1}^n i \cdot x_i^2$	[-100,100]	0

Table 2 Test Multimodal Functions

Name	Function	range	f_{min}
Alpine	$F_8(x) = \sum_{i=1}^n x_i \cdot \sin(x_i) + 0.1x_i $	[-100,100]	0
Griewank	$F_9(x) = \frac{1}{4000} \sum_{i=1}^n x_i^2 - \prod_{i=1}^n \cos\left(\frac{x_i}{\sqrt{i}}\right) + 1$	[-100,100]	0
Rastrigin	$F_{10}(x) = \sum_{i=1}^n (x_i^2 - 10 \cos(2\pi x_i) + 10)$	[-100,100]	0
Ackley	$F_{11}(x) = 20 + e - 20 \exp\left(-0.2 \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2}\right) - \exp\left(\frac{1}{n} \sum_{i=1}^n \cos(2\pi x_i)\right)$	[-100,100]	0
Lévy	$F_{12}(x) = \sin^2(\pi w_1) + \left(\sum_{i=1}^{n-1} (w_i - 1)^2 (1 + 10 \sin^2(\pi w_i + 1))\right) + (w_n - 1)^2 (1 + \sin^2(2\pi w_n))$, $w_i = 1 + \frac{x_i - 1}{4}$	[-100,100]	0
Salomon	$F_{13}(x) = 1 - \cos(2\pi \sqrt{\sum_{i=1}^n x_i^2}) + 0.1 \sqrt{\sum_{i=1}^n x_i^2}$	[-100,100]	0
Schaffer	$F_{14}(x) = 0.5 + \frac{1}{D-1} \sum_{i=1}^{D-1} (\sin(\sqrt{x_i^2 + x_{i+1}^2})^2 - 0.5)$	[-100,100]	0

5.2 Trend Analysis

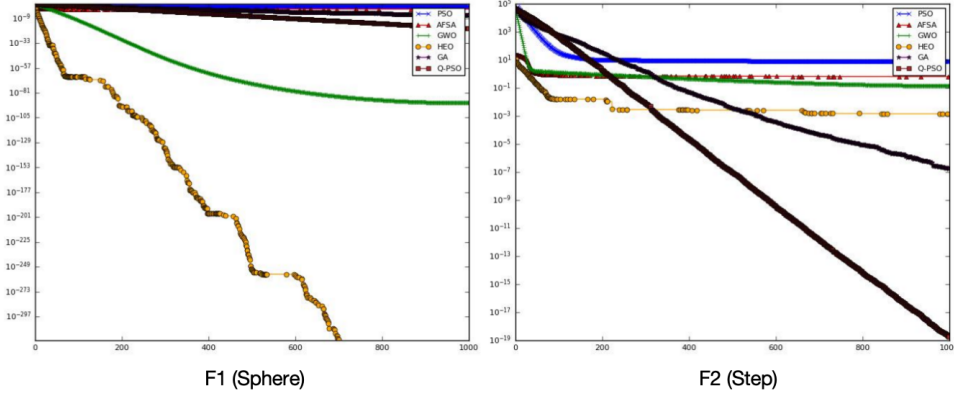


Fig. 5 The cost variation in F_1 and F_2

In Fig.5, the HEO achieves 0 costs and converges faster than other algorithms in F_1 . The GWO gets the second low-cost. As Table 3 indicates, HEO gets the 0 cost and GWO gets the $1.651\text{e-}91$ cost, which could also be considered to be almost the global optimum.

The Fig.5 demonstrates the QPSO gets the smallest cost among the others, and the GA and HEO are next in F_2 . As for the convergence, except QPSO, the HEO has get relatively low cost in the beginning to about the 600 iterations. After that, the HEO's performance is exceeded by the GA. Also, the slope of the GA plot infers the acceptable convergence in further searching, it infers the single constraint conditions might make the HEO hard to converge due to the frequent skip and vibration.

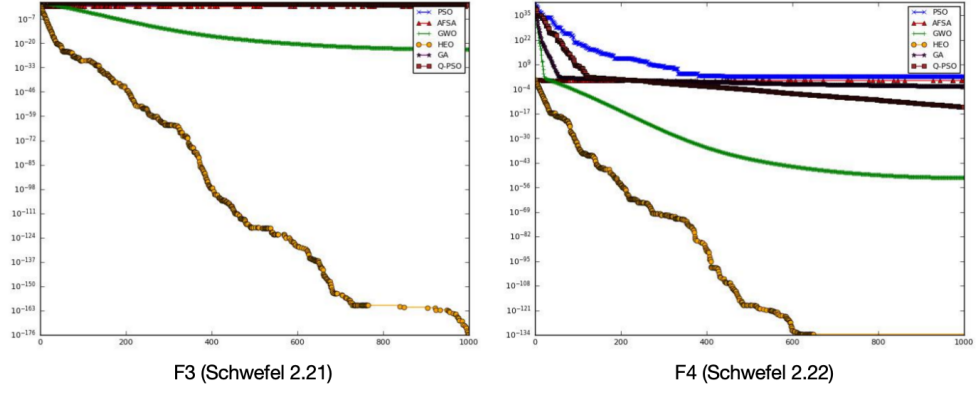


Fig. 6 The cost variation in F_3 and F_4

For F_3 , the Fig.6 shows the fast convergence of HEO. Other algorithms including GWO seem only to find local optima of the function. Even the cost of GWO is $6.178e-24$ (seen in Table 3), which cannot be seen as an ideal situation.

For F_4 , Fig.6 also shows that the performances of HEO are much superior to GA, AFSA, and PSO. The HEO only gets around $1e-134$ cost, and GWO achieves about $1e-50$. The QPSO, AFSA, GWO, and GA express a similar trait on the plot, those algorithms all converge smoothly but stuck in optimums. Adversely, the oscillations of PSO could be caused by the complex landscape of the F_4 function, and the fluctuations of the HEO plot could be caused by its unique random mechanism, such as random skip and its escape behavior.

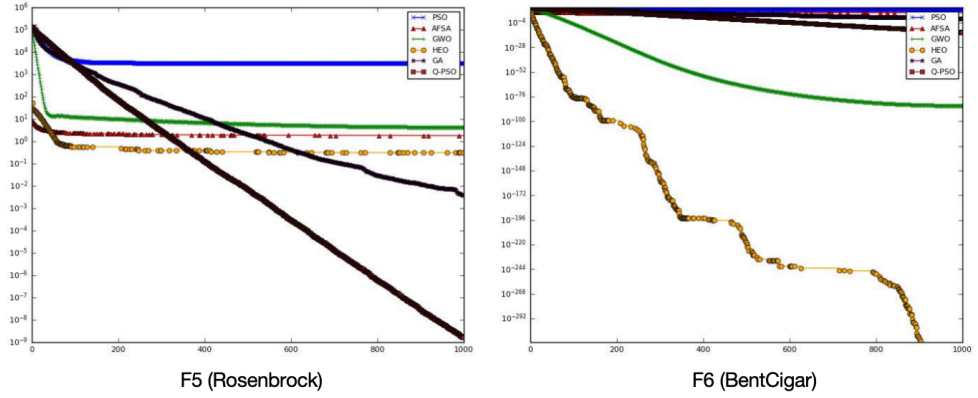


Fig. 7 The cost variation in F_5 and F_6

For F_5 [Rosenbrock \(1960\)](#), Fig.7 shows the majority of algorithms get trapped by the special curve structure, and only GA as well as QPSO gets a good convergence. Besides, the HEO also shows a reasonable cost that is lower than the others.

For F_6 , Fig.7 also shows the HEO finds the global optima, and the GWO algorithm also finds a good solution, as in Table 3, we can observe a cost of $1.893e-85$ cost. Other algorithms except GA do not have apparent tendency and they all converge too early as the plot shows.

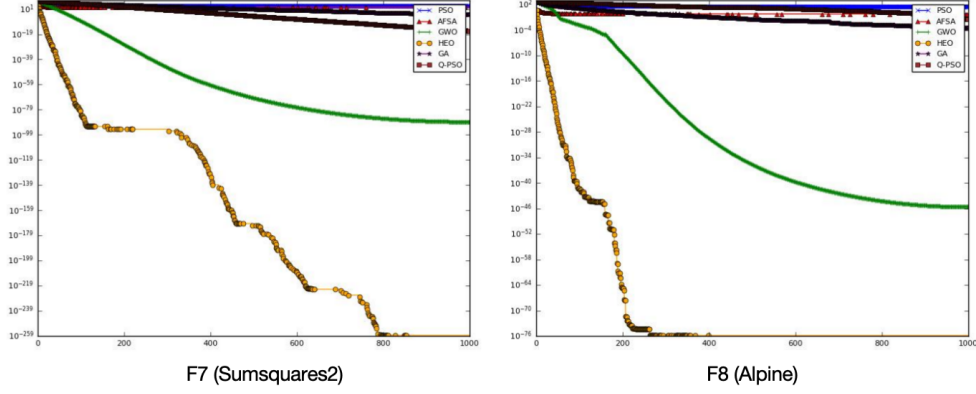


Fig. 8 The cost variation in F_7 and F_8

For F_7 , the Fig.8 shows the HEO gets a good solution, and GWO follows after it. As for the convergence, HEO converges with the fastest rate, and GWO gets the second fastest rate, notwithstanding the AFSA getting the smallest cost in the first few iterations.

For F_8 , Fig.8 also shows the HEO gets the best performance compared to other algorithms, and GWO gets the second-best performance at around $1e-45$ cost. Unlike HEO performs in other functions, the HEO gets a gentle plot shape, which might caused by the smooth landscape of the F_8 .

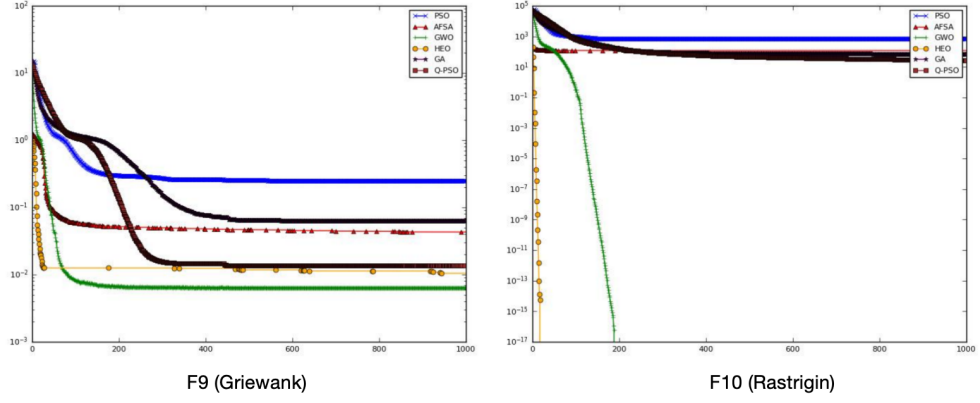


Fig. 9 The cost variation in F_9 and F_{10}

For F_9 Griewank (1981), Fig.9 illustrates all algorithms that encounter the difficulty of finding The global optima. The F_9 , Griewank, is a complex function. With $p=30$, the algorithms are easy to trap in a single convex hull that does not contain the best solution. The HEO and GWO get approximate costs, but HEO converge faster and get slightly lower costs.

For F_{10} , Fig.9 also shows that GWO as well as HEO both find the global optimum. For analyzing the convergence, the HEO finds the solution faster than the GWO.

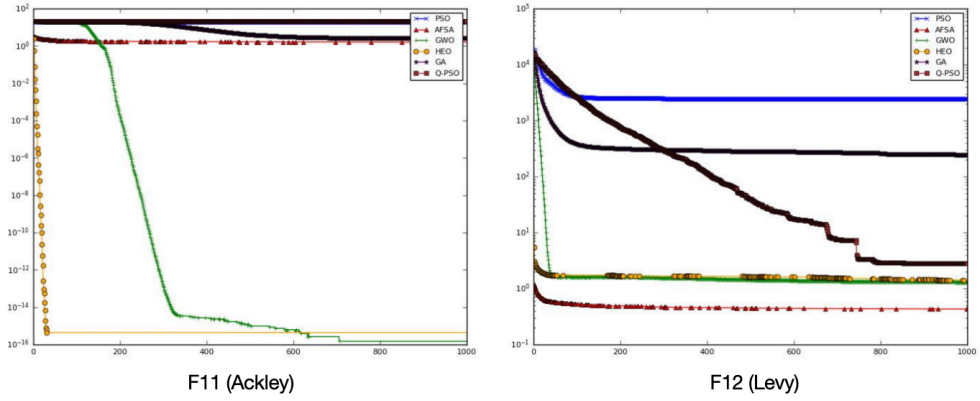


Fig. 10 The cost variation in F_{11} and F_{12}

For F_{11} , Fig.10 shows that HEO and GWO both obtain a reasonable cost around $1e-16$, but do not get full convergence. The possible reason for explaining this result could be the hollows of the Ackley function F_{11} in high dimension Cai et al (2020).

Those hollow landscapes with the same cost might be the issue that prevents further convergence of algorithms.

For F_{12} , Fig.10 also shows that the AFSA gets the lowest cost and then follows with GWO and HEO. All testing algorithms, including AFSA, do not perform well. The Lévy functions is a folding bent shape function with some sheer landscapes, which might indicate a drawback for HEO.

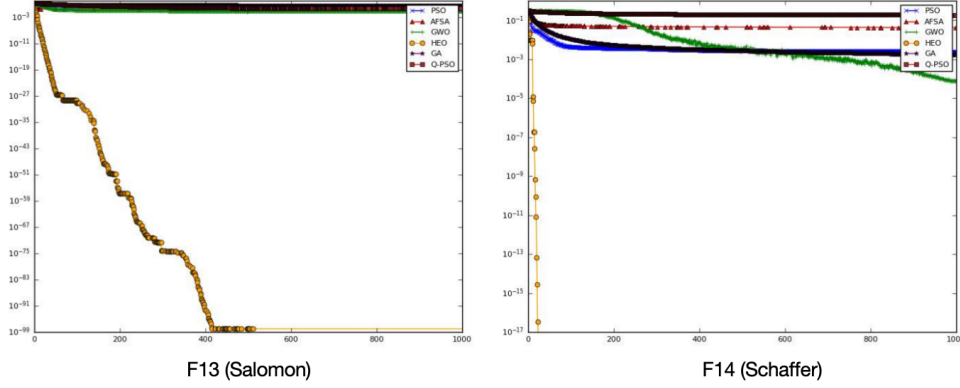


Fig. 11 The cost variation in F_{13} and F_{14}

For F_{13} , Fig.11 shows the HEO gets the best result, which is near the global optimum. Salomon function F_{13} is another sheer function but with an extent of convexity, making HEO able to find the solution.

For F_{14} , Fig.11 also shows the HEO found the best solution in the first few iterations. In addition, the GWO has a small inclination to converge further and possibly to find the global optima.

In summary, from analysis of 14 benchmark functions, illustrates Halfway Escape Optimization (HEO) gets successful convergences on each case, which validates with the effectiveness of HEO.

5.3 Experimental Comparison

The bold text in Table 3 is the smallest cost for one objective function. As the overall result that Table 3 shows, the HEO gets a better comprehensive performance than PSO, AFSA, GWO, GA, and QPSO in major benchmark functions F_1 to F_{14} , and get the minimum costs over 9 functions. On another side, HEO gets a relatively bigger cost in F_2 , F_5 , and F_{12} , which might infer the disadvantages of using escape and skip mechanisms in HEO.

For a more precise analysis, the article performs a straightforward rank aggregation procedure on the data presented in Table 3. Algorithms with lower associated costs are accorded higher ranks within this aggregation. The resulting aggregated rank data is displayed in Table 4. In Table 4, Halfway Escape Optimization (HEO) achieves the

Table 3 Mean Costs of the Algorithms

	PSO	AFSA	GWO	HEO	GA	QPSO
F_1	3.545e+02	7.847e-01	1.651e-91	0.000e+00	8.814e-07	1.465e-19
F_2	7.943e+00	7.992e-01	1.416e-01	1.344e-03	1.742e-07	1.880e-19
F_3	1.420e+01	4.495e-01	6.178e-24	1.302e-176	5.579e+00	2.303e+00
F_4	8.246e+02	3.912e+00	1.622e-51	2.498e-134	4.592e-03	3.713e-14
F_5	3.168e+03	2.235e+00	4.271e+00	3.112e-01	3.817e-03	1.603e-09
F_6	1.022e+09	8.106e+05	1.893e-85	0.000e+00	1.077e+00	6.314e-14
F_7	2.470e+03	3.790e+02	2.168e-89	3.531e-259	4.843e-04	4.084e-17
F_8	3.381e+01	7.514e-01	2.441e-46	1.299e-76	3.697e-04	4.062e-02
F_9	0.247727	0.053917	0.006319	0.010466	0.064001	0.013684
F_{10}	715.590208	137.192018	0.000000	0.000000	67.088411	27.315077
F_{11}	2.000e+01	1.811e+00	1.480e-16	4.440e-16	2.665e+00	2.088e+01
F_{12}	2439.240184	0.544817	1.314077	1.421046	245.547882	2.834483
F_{13}	4.046e+00	2.954e-01	5.659e-02	9.341e-99	3.357e+00	3.932e-01
F_{14}	0.002472	0.050465	0.000076	0.000000	0.001709	0.186564

Table 4 The Average Ranks of the Costs

	PSO	AFSA	GWO	HEO	GA	QPSO
Unimodal Functions	5.7142	4.5714	2.7142	1.5714	3.7142	2.7142
Multimodal Functions	5.4285	3.5714	1.5714	1.5714	4.0000	4.1428
Total Rank($F_1 - F_{14}$)	5.5714	4.0714	2.1428	1.5714	3.8571	3.4285

highest rank in both unimodal functions (encompassing $F_1 - F_7$) and multimodal functions (encompassing $F_8 - F_{14}$). This suggests that HEO exhibits similar performance compared to the Grey Wolf Optimizer (GWO) specifically in handling multimodal Functions, it also demonstrates competitive performance relative to the other six algorithms tested in unimodal functions, even in scenarios involving relatively complex functions.

Table 5 Search Time(s/1000 iters) of the Algorithms

	PSO	AFSA	GWO	HEO	GA	QPSO
F_1	2.732496	136.057096	74.678300	18.509185	9.189708	25.709296
F_2	3.307250	150.451374	76.191960	20.107816	10.300027	26.042222
F_3	0.522904	36.712690	34.066304	8.800284	4.023997	10.214057
F_4	1.587224	86.004913	72.936023	15.730473	8.095256	24.084705
F_5	6.240265	237.248756	79.063583	25.920876	13.020591	29.193232
F_6	2.916587	139.673755	75.215040	18.904487	9.627195	25.440708
F_7	15.078621	509.393016	87.199845	43.967146	21.811757	37.721174
F_8	1.059602	63.582046	43.069062	12.031251	6.521126	13.315242
F_9	10.000640	315.942743	53.234886	31.335585	15.717625	22.645855
F_{10}	10.626591	309.697071	83.785108	35.644457	17.775824	33.821670
F_{11}	12.349613	381.549480	85.610975	38.228737	19.336686	34.786341
F_{12}	7.995789	266.661428	50.095542	27.491028	14.171460	20.122009
F_{13}	3.320846	45.687899	77.897484	19.997064	10.503721	26.490371
F_{14}	7.923606	273.046009	51.550709	26.627504	14.288681	20.465703

For the efficiency, based on the data from Table 5, the search time of five algorithms in these functions always shows a relationship that: $T_{PSO} < T_{GA} < T_{HEO} \approx T_{QPSO} < T_{GWO} \approx T_{AFSA}$. In conclusion, the HEO demonstrates an acceptable search time as well as a good performance among test functions except for F_{12} , but further validation of its effectiveness is still needed. For example, for a more comprehensive view, further tests could focus on testing HEO in other objective functions and various dimensions instead of just 30. This broadened experiment could give a more precise evaluation of the HEO.

6 Stimulated Engineering Problems

6.1 The Pressure Vessel Design Problem

The pressure vessel design is a classical engineering problem, the objective is to change the structure of the vessel to reduce its manufacturing cost [Sandgren \(1990\)](#). The optimization algorithm tested in this problem could well validate its performance in complex constrained situations. As Fig.12 and Fig.13 shown, the vessel composed by a cylinder shape of the structure welded with a sphere bottom. By changing from x_1 to x_4 values (x_1 : (R -Radius of the Shell), x_2 : (L -Length of the Shell), x_3 : (T_s -Thickness of the Shell), x_4 : (T -Thickness of the dish end)), the task is to minimize the cost in equation(18).

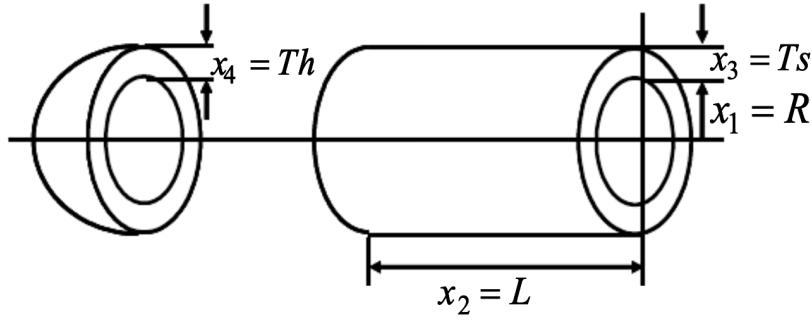


Fig. 12 The Structure of the Vessel-[Jacob \(2014\)](#)

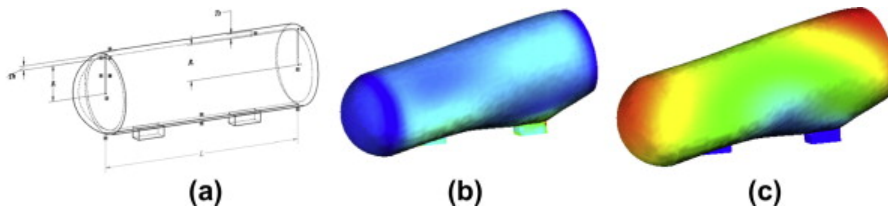


Fig. 13 The Heatmaps of the Vessel-[Mirjalili et al \(2014\)](#)

Consider :

$$\tilde{\mathbf{x}} = [x_1, x_2, x_3, x_4]$$

Minimize :

$$f(\tilde{\mathbf{x}}) = 0.6224x_1x_2x_3 + 1.7781x_1^2x_3 + 3.1161x_2x_4^2 + 19.84x_4x_1^2$$

Subject to :

$$g_1(\tilde{\mathbf{x}}) = -x_1 + 0.0193x_3 \leq 0$$

$$g_2(\tilde{\mathbf{x}}) = -x_3 + 0.000954x_3 \leq 0 \quad (18)$$

$$g_3(\tilde{\mathbf{x}}) = -\pi x_3^2x_4 + \frac{4}{3}\pi x_3^3 + 1296000 \leq 0$$

$$g_4(\tilde{\mathbf{x}}) = x_4 - 240 \leq 0$$

Variable Range :

$$0.00625 \leq x_1 \leq 1.25$$

$$0.00625 \leq x_2 \leq 1.25$$

$$40 \leq x_3 \leq 200$$

$$40 \leq x_4 \leq 200$$

Table 6 Pressure Vessel Design in 30 times Validation

	PSO	GWO	HEO	GA	QPSO
μ_{cost}	9938.35	9197.10	8638.74	8757.88	7680.54
σ_{cost}	1952.30	1890.72	1268.32	1533.92	825.02
T_s^*	0.8125	1.25	1.25	1.0	0.9375
T_h^*	0.4375	0.5625	0.5625	0.5	0.5
R^*	40.3227	58.9299	57.6567	51.4577	88.9923
L^*	200.0	40.4031	47.4311	48.3081	112.6784

As Table 6 and Fig.14 illustrated, the HEO algorithm show a relativelyx small cost among the others, but larger than the cost of QPSO. The Pressure Vessel Design Problem is a linear combination problem with constraints, which is similar to the case of the Step Function. From the HEO's cost in the Step Function(or F_2) being larger than QPSO,it is not unexpected that HEO perform worse than QPSO in this problem, which might indicate HEO's weakness when solving the complex problem with simple constraints.

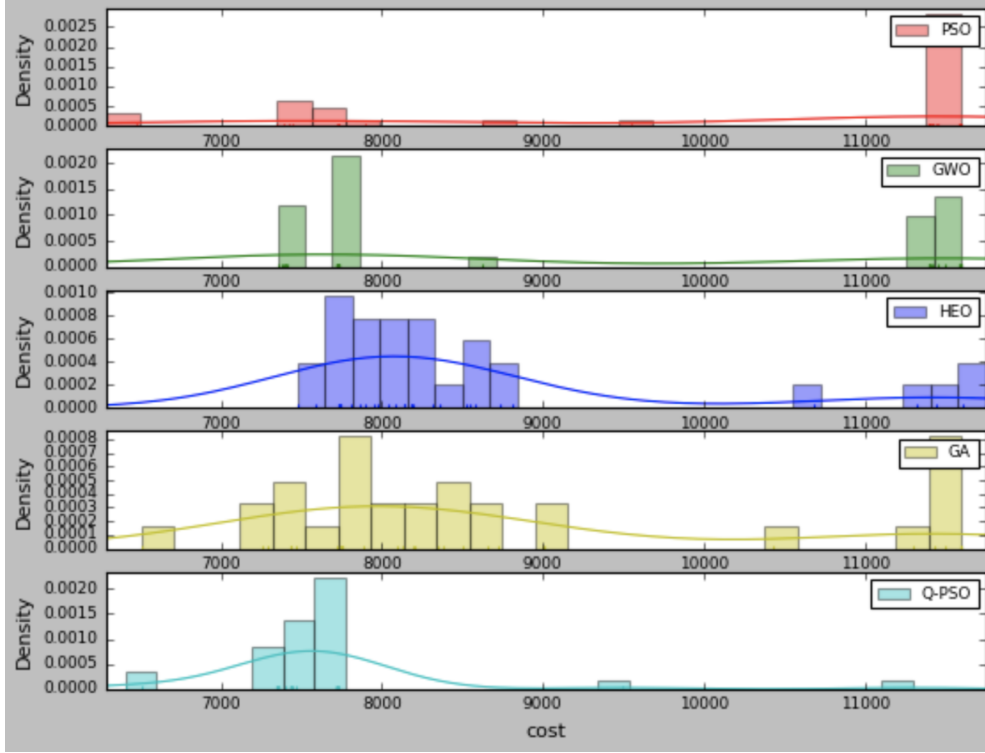


Fig. 14 The Distribution of Cost for Pressure Vessel Design

6.2 Tubular Column Design Problem

As Fig.15 shown, the Tubular Column this problem is structural part that is a cylindrical shaft, which may be solid, or made of a single circular cross-section and produced from metal, concrete, and other materials [Tub \(2018\)](#). It is commonly used within constructing projects where it is required to support beams, as well as other elements, such as columns, and so on, and it is also used in bridges and other types of structures.

Typically, tubular columns are superior and efficient compared to solid columns because they can resist torsion, bending, and shear force. The ideal is to minimize the cost of realising the column which is captured by the variables d , the average diameter of the column in centimeters and t the thickness of the column in centimeters.

In Table 7, the paper sets 5 algorithms with the same parameter as the previous tests in the experiment (HEO with $c_{max} = 1$), the AFSA does not show any inclination to converge to test engineering problems; hence, relevant data are eliminated in the table. The HEO has the smallest mean value of cost compared to other algorithms in 30 times validations, might illustrate its good performance in the tubular column problem, or relatively simple function with complex constraints. As for the deviations, the GA get the smallest one, indicating its stability to some extent.

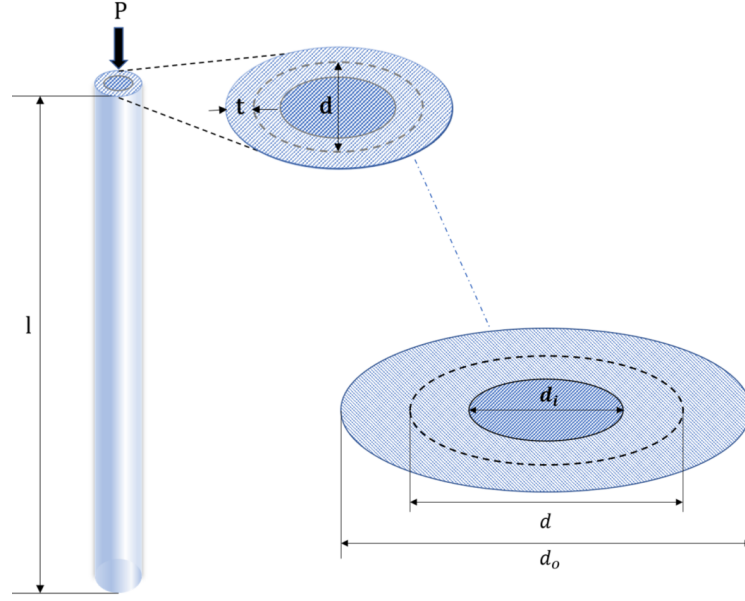


Fig. 15 The Structure of the Tubular Column Design-Tub (2018)

Consider :

$$\tilde{\mathbf{x}} = [x_1, x_2]$$

Minimize :

$$f(\tilde{\mathbf{x}}) = 9.8x_1x_2 + 2x_1$$

Subject to :

$$g_1(\tilde{\mathbf{x}}) = \frac{P}{\pi x_1 x_2 \chi_y} - 1 \leq 0$$

$$g_2(\tilde{\mathbf{x}}) = \frac{8PL^2}{\pi^3 E x_1 x_2 (x_1^2 + x_2^2)} - 1 \leq 0$$

$$g_3(\tilde{\mathbf{x}}) = \frac{2}{x_1} - 1 \leq 0$$

$$g_4(\tilde{\mathbf{x}}) = \frac{x_1}{14} - 1 \leq 0$$

$$g_5(\tilde{\mathbf{x}}) = \frac{0.2}{x_2} - 1 \leq 0$$

$$g_6(\tilde{\mathbf{x}}) = \frac{x_2}{8} - 1 \leq 0$$

Variable Range :

$$2 \leq x_1 \leq 14$$

$$0.2 \leq x_2 \leq 0.8$$

(19)

Table 7 Tubular Column Design in 30 times Validation

	PSO	GWO	HEO	GA	QPSO
μ_{cost}	32.105096	32.147502	31.704160	32.224944	32.144308
σ_{cost}	0.419599	0.366717	0.760805	0.015721	0.369621
d^*	7.1031	7.1013	7.1477	8.1346	7.1013
t^*	0.229	0.2295	0.2279	0.2	0.2292

Table 8 Zero-minimum benchmark results (30D, 100 particles, 1000 iterations, 30 runs). Lower is better.

Function	Original HEO	v2 (ES 1/5)	CIBA	DirCIBA
Sphere	2.87×10^4	4.23×10^1	2.96×10^2	1.70×10^1
Rosenbrock	1.03×10^{10}	1.08×10^4	1.95×10^6	3.60×10^3
Rastrigin	2.99×10^4	2.97×10^2	4.85×10^2	2.63×10^2
Griewank	8.17×10^0	9.23×10^{-1}	7.39×10^{-1}	7.56×10^{-1}
Levy	8.11×10^3	7.36×10^0	1.40×10^2	8.87×10^0

Table 9 Non-zero benchmark results (shifted+offset). Known global minimum is $c = 123.456$. Lower gap is better.

Function	v2 mean	v2 gap	DirCIBA mean	DirCIBA gap
Shifted Sphere	2735.10	2611.64	2232.97	2109.51
Shifted Rastrigin	2502.98	2379.53	2207.81	2084.36
Shifted Rosenbrock	25567.77	25444.31	15334.48	15211.02
Shifted Griewank	124.85	1.40	124.83	1.38
Shifted Levy	745.45	621.99	896.43	772.98

Among them in equation(19), $P = 2300$ means compressive load with unit kg_f , $chi_y = 450$ (is yield stress, has unit kg_f/cm^2), $E = 0.65 * 10^6$ is elasticity, unit is kg_f/cm^2), as for $L = 300$, is length of column with unit cm .

6.3 Zero-minimum Benchmarks (Mean $\pm$ Std)

Table 8 reports the mean and standard deviation of the best objective value over 30 independent runs for selected zero-minimum benchmark functions in 30 dimensions.

6.4 Non-zero Benchmarks (Shifted + Offset)

To evaluate robustness beyond zero-centered objectives, we follow a shifted+offset protocol:

$$f_{\text{shift}}(x) = f(x - s) + c, \quad (13)$$

where $c > 0$ ensures a non-zero global minimum and s shifts the optimum away from the origin.

Table 9 reports the mean best objective values and the mean gap to the known optimum value (c).

7 Model-based Optimization for Real-time Problem

The experimental results indicate that HEO outperforms other algorithms in terms of convergence speed and solution quality across a range of benchmark functions. The algorithm's ability to balance exploration and exploitation, coupled with its adaptive strategies, has resulted in competitive performance in challenging optimization landscapes. The discussion of stimulated engineering problems like Tubular Column Design Problem highlights the effectiveness of the HEO algorithm in general applications.

However, whether the performance of the HEO is enough to deal with real-time situations still questionable, the applications of HEO may have to be more focused on realms like industrial parts manufacturing instead of a more dynamic environment. In addition, some special functions like Lévy or Step would still be an issue for HEO to explore solutions, the Pressure Vessel Design Problem also infer its shortage.

For further testing of the HEO, the paper would use the tested optimization algorithms and Grid Search(because it is a good baseline for hyperparameter tuning [Leon and Xiong \(2015\)](#)) with 50 population size and 50 iterations(2500 size sample space for Grid Search) to optimize the Logistic Regression parameters for better performance in this classification data set, due to the Logistic Regression do not have any randomness(The definition of the Logistic Regression had shown in equation (20)) [Cox \(1958\)](#). The first parameter is the Cost Parameter C in Logistic Regression, and the second one is the maximum iteration i_{max} to minimize the mean square error shown in equation (21) of the model. This method that find candidate solutions for model's parameter directly called the Model-based Optimization [Chen and Zhou \(2015\)](#), Model-based Optimization provides a controlled and repeatable framework for testing, ensuring consistent and reliable results.

$$\begin{aligned}
 h_{\theta}(\mathbf{x}) &= \frac{1}{1 + \exp(-\theta^T \mathbf{x})} \\
 J(\theta) &= -\frac{1}{m} \sum_{i=1}^m [\mathbf{y}^{(i)} \ln(h_{\theta}(\mathbf{x}^{(i)})) + (1 - \mathbf{y}^{(i)}) \ln(1 - h_{\theta}(\mathbf{x}^{(i)}))] \\
 \frac{\partial J(\theta)}{\partial \theta_j} &= \frac{1}{m} \sum_{i=1}^m (h_{\theta}(\mathbf{x}^{(i)}) - \mathbf{y}^{(i)}) \mathbf{x}_j^{(i)}
 \end{aligned} \tag{20}$$

Consider :

$$\tilde{\mathbf{x}} = [x_1, x_2]$$

Minimize :

$$f(\tilde{\mathbf{x}}) = \sum \frac{((\beta_1 x_1 + \beta_2 x_2) - y)^2}{2} \tag{21}$$

Variable Range :

$$1e - 16 \leq x_1 \leq 100$$

$$1 \leq x_2 \leq 100$$

Rice industry is world-wide and have a demand to selecting out the ones that satisfy the various quality criteria of countries [Soe et al \(2015\)](#). The paper by Kay CINAR and Murat KOKLU suggest a new tabular dataset extracted from image characteristics to identify two rice species Osmancik-97 and Cammeo [Cinar and Koklu \(2019\)](#), a binary classification contains 2180 samples of Osmancik and 1630 for Cammeo. The performance of the models shown in the Table 10, with the same ratio(test set:train set is 0.25:0.75), the best model with HEO has a performance(accuracy: 0.9328, f1-score: 0.9328) that exceeds the record (accuracy: 0.9302, f1-score: 0.9180) in the original paper. The Evaluation Metrics including:

$$\begin{aligned}
\text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN} \\
\text{Sensitivity} &= \frac{TP}{TP + FN} \\
\text{Specificity} &= \frac{TN}{TN + FP} \\
\text{Precision} &= \frac{TP}{TP + FP} \\
\text{Recall} &= \frac{TP}{TP + FN} \\
\text{F1-score} &= 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}
\end{aligned} \tag{22}$$

Where:

- TP represents the number of true positives.
- TN represents the number of true negatives.
- FP represents the number of false positives.
- FN represents the number of false negatives.

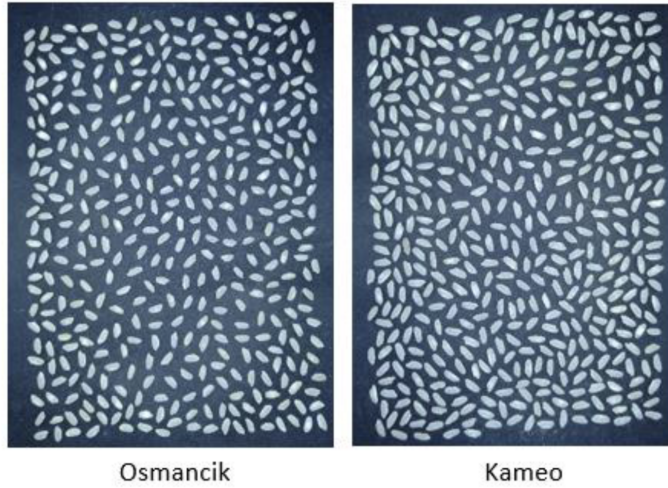


Fig. 16 The raw sample of the Rice Dataset-[Cinar and Koklu \(2019\)](#)

Table 10 Rice Classification in Test Set

	PSO	GWO	HEO	GA	QPSO	GridSearch
Accuracy	0.9317	0.9296	0.9328	0.9317	0.9296	0.9296
Sensitivity	0.9418	0.9381	0.9418	0.9418	0.9381	0.9381
Specificity	0.9181	0.9181	0.9205	0.9181	0.9181	0.9181
Precision	0.9317	0.9296	0.9328	0.9317	0.9296	0.9296
Recall	0.9317	0.9296	0.9328	0.9317	0.9296	0.9296
F1-score	0.9317	0.9296	0.9328	0.9317	0.9296	0.9296
SearchTime(s/it)	119.76	138.67	79.32	140.98	120.29	22.28
$x_1(C)$	15.8281	11.6165	7.2574	7.6160	15.2641	6.0000
$x_2(i_{max})$	60	74	77	62	74	51

8 Conclusion

Through its quantum-inspired metaheuristic, Halfway Escape Optimization (HEO) demonstrates adaptability, efficiency, and effectiveness in tested problems. The algorithm’s unique energy-driven behavior, vibration strategies, and exploratory mechanisms contribute to its ability to balance exploration and exploitation, leading to fast convergence rates and high-quality solutions.

Moreover, the adaptability of the HEO algorithm allows it to be applied to dynamic environments where the objective functions or constraints may change over time. By continuously adapting its search strategy and updating the position of particles, HEO can effectively track and respond to changing optimization landscapes in different engineering problems, ensuring that high quality solutions are maintained even in dynamic scenarios.

Overall, the HEO has demonstrated its stable effectiveness in 14 benchmarks functions and shown adaptability in addressing general optimization like the Pressure Vessel Design Problem, the Tubular Column Design Problem and the Model-based Optimization Problem. Future work could focus on further refining the algorithm’s parameters and exploring its application in other practical optimization problems like path planning [Tanakitkorn et al \(2014\)](#), circuit planning [Saw et al \(2023\)](#). The algorithm could also be extended to address multi-objective optimization problems and dynamic environments or change the strategy of escape and skip, further enhancing its versatility and applicability.

Conclusion (Extended Study).

In addition to reproducing the original HEO algorithm, we introduced a set of extensions aimed at improving stability and convergence. First, we removed the stochastic amplifier term r_2 from the position update and introduced a bounded escape multiplier to control the effect of stagnation. Second, we replaced the original vibration mechanism with an annealed noise schedule that vanishes over time, enabling late-stage contraction. Third, we integrated a $(1 + 1)$ ES local search controlled by the $1/5$ success rule (v2), which significantly improved convergence on unimodal zero-minimum benchmarks. We also investigated immune-inspired affinity mutation (CIBA) and a

directional variant (DirCIBA), where directional guidance toward the best solution mitigates random-walk behavior near optima. While the proposed variants substantially improve performance on standard zero-minimum benchmarks, shifted+offset experiments show that all variants degrade when the optimum is displaced and the minimum is non-zero, highlighting the importance of evaluating escape-based metaheuristics on non-zero benchmark protocols.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRedit authorship contribution statement

Jiawen Li: Conceptualization, Methodology, Writing-Original Draft, Writing-Editing. **Anwar PP Abdul Majeed:** Supervision, Writing-Review, Validation. **Pascal Lefevre:** Supervision, Writing-Review, Validation.

Data availability

The experiment is data-available for replications, provide the source code of HEO in: <https://github.com/Spike8086/HEO.git>

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